

Fake News Detector AI: A Machine Learning-Based System for Detecting Misinformation in News Articles

Authors: Kiera & Omar

Course: Artificial Intelligence Project

Institution: University of The Gambia

Academic Year: 2025/2026

Abstract

The rapid spread of misinformation through online platforms has become a significant global challenge. Fake news can influence public opinion, create confusion, and contribute to the spread of inaccurate information. This project presents the development of a Fake News Detector AI system that automatically classifies news articles as real or fake using Machine Learning and Natural Language Processing (NLP) techniques. The system was built using Python, Scikit-learn, Pandas, and Streamlit. A dataset consisting of real and fake news articles was collected and preprocessed before training a Logistic Regression model. TF-IDF vectorization was used to convert textual data into numerical features suitable for machine learning. The trained model was integrated into a Streamlit web application, allowing users to enter news articles and receive predictions in real time. The project demonstrates the complete AI development lifecycle, including problem identification, data preprocessing, model training, evaluation, deployment, and collaborative development using Git and GitHub. The resulting system provides a practical demonstration of how artificial intelligence can assist in combating misinformation.

Keywords: Fake News Detection, Machine Learning, Natural Language Processing, Logistic Regression, Streamlit, Artificial Intelligence.

1. Introduction

The internet has revolutionized the way information is shared and consumed. News can now reach millions of people within seconds through websites, blogs, and social media platforms. While this accessibility has many benefits, it has also increased the spread of fake news and misinformation.

Fake news refers to false or misleading information presented as legitimate news. Such information can influence public opinion, affect elections, spread panic, and damage reputations. Detecting fake news manually is difficult because of the enormous amount of content published daily.

Artificial Intelligence and Machine Learning provide an opportunity to automate the process of fake news detection. By analyzing patterns in text data, machine learning models can learn to distinguish between real and fake news articles.

The objective of this project is to design, train, and deploy an AI-driven Fake News Detector capable of classifying news articles accurately while providing a user-friendly interface for practical use.

2. Problem Statement

The increasing spread of misinformation poses serious social, political, and economic challenges. Traditional methods of fact-checking are time-consuming and require significant human effort.

The problem addressed by this project is:

How can machine learning be used to automatically classify news articles as real or fake in an efficient and accessible manner?

The project aims to provide a practical solution that assists users in identifying potentially misleading information.

3. Objectives

General Objective

To develop an AI-powered fake news detection system capable of classifying news articles as real or fake.

Specific Objectives

- Collect and prepare a dataset containing real and fake news articles.
 - Apply Natural Language Processing techniques to clean and transform textual data.
 - Train a machine learning model for classification.
 - Evaluate the model's performance.
 - Deploy the model using a Streamlit web application.
 - Demonstrate collaborative software development using Git and GitHub.
-

4. Related Work / Literature Review

Fake news detection has become an important research area within Natural Language Processing and Machine Learning.

Several approaches have been proposed:

Rule-Based Systems

Early fake news detection systems relied on manually defined rules and expert knowledge. While useful, these systems lacked flexibility and scalability.

Machine Learning Approaches

Traditional machine learning algorithms such as:

- Logistic Regression
- Naive Bayes
- Support Vector Machines (SVM)

have been widely used for text classification tasks due to their efficiency and interpretability.

Deep Learning Approaches

Recent studies have explored:

- Recurrent Neural Networks (RNN)
- Long Short-Term Memory (LSTM)
- BERT Transformers

These models often achieve higher accuracy but require larger datasets and greater computational resources.

This project focuses on Logistic Regression because it offers a strong balance between performance, simplicity, and computational efficiency for academic projects.

5. Methodology

5.1 Data Collection

The project used publicly available datasets:

- Fake.csv
- True.csv

These datasets contain news articles labeled as either fake or real.

The datasets were combined into a single dataset for training and testing purposes.

5.2 Data Preprocessing

Before training the model, the text data was cleaned using Natural Language Processing techniques.

The preprocessing steps included:

- Converting text to lowercase
- Removing URLs
- Removing special characters
- Removing unwanted symbols
- Standardizing text format

These steps help improve model performance by reducing noise in the data.

5.3 Feature Extraction

Machine learning algorithms cannot directly process raw text.

Therefore, the project used:

TF-IDF Vectorization

Term Frequency–Inverse Document Frequency (TF-IDF) converts textual information into numerical vectors.

Advantages:

- Highlights important words
 - Reduces influence of common words
 - Improves classification performance
-

5.4 Model Development

The selected machine learning algorithm was:

Logistic Regression

Reasons for selection:

- Effective for binary classification
- Computationally efficient

- Easy to interpret
- Suitable for text classification tasks

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

The model was trained using Scikit-learn.

5.5 Model Saving

After training, the model and vectorizer were saved using Joblib:

- model.pkl
- vectorizer.pkl

These files allow predictions without retraining the model.

6. System Architecture

The system architecture follows the workflow below:

1. User enters news article.
2. Streamlit interface captures input.
3. Text preprocessing occurs.
4. TF-IDF vectorizer transforms text.
5. Logistic Regression model predicts class.
6. Result displayed to user.

Components

Frontend

- Streamlit

Backend

- Scikit-learn
- Joblib

Data Processing

- Pandas
- NumPy

Version Control

- Git
 - GitHub
-

7. Deployment

The project was deployed as a Streamlit web application.

Users interact with the system through a graphical interface where they can:

- Paste news articles
- Analyze content
- Receive classification results
- View confidence scores

The deployment demonstrates how machine learning models can be integrated into real-world applications.

8. Experiments and Results

The model was trained and evaluated using the prepared dataset.

Evaluation metrics included:

- Accuracy

- Confusion Matrix
- Classification Report

The trained Logistic Regression model successfully classified news articles as real or fake.

The system demonstrated effective performance for educational and demonstration purposes.

Sample Predictions

Input	Prediction
Government announces education reform	Real
Scientists confirm humans can become invisible	Fake
WHO releases health guidelines	Real
Aliens secretly control governments	Fake

9. Discussion

The project demonstrates the practical application of machine learning for fake news detection.

Strengths

- User-friendly interface
- Fast prediction speed
- Efficient machine learning pipeline
- Easy deployment using Streamlit

Limitations

- Depends on training data quality
- Cannot guarantee factual correctness

- May struggle with unseen writing styles

Despite these limitations, the project successfully demonstrates the feasibility of AI-based fake news detection.

10. Challenges Encountered

Several challenges were faced during development:

- Integrating frontend and backend components
- Managing Git branches and merge conflicts
- Loading trained models correctly
- Handling dependency installation issues

These challenges provided valuable experience in software engineering and team collaboration.

11. Future Improvements

Future enhancements may include:

- Deep learning models (BERT, LSTM)
 - Real-time news scraping
 - API integration using FastAPI
 - Cloud deployment
 - Larger datasets
 - Multi-language support
-

12. Conclusion

This project successfully developed a complete AI-driven Fake News Detector capable of classifying news articles as real or fake. The project

covered all stages of the AI lifecycle, including data collection, preprocessing, model training, evaluation, deployment, and integration.

The resulting system demonstrates how Machine Learning and Natural Language Processing can be applied to address the real-world challenge of misinformation. Additionally, the project provided practical experience in collaborative software development using Git and GitHub.

References

1. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*.
2. Bird, S., Klein, E., & Loper, E. (2009). *Natural Language Processing with Python*. O'Reilly Media.
3. Jurafsky, D., & Martin, J. H. (2023). *Speech and Language Processing*.
4. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
5. Python Software Foundation. Python Documentation. <https://www.python.org>
6. Scikit-learn Documentation. <https://scikit-learn.org>
7. Streamlit Documentation. <https://streamlit.io>